

Flood Risk Assessment for the UQ St. Lucia Campus – Summary Report

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A report prepared for UQ Property and Facilities to assess flood risk in order to manage and prevent potential future losses. The University is located on a magnificent site of 114-hectares bounded on three sides by the Brisbane River, but the river is its greatest hazard when it floods. This map shows the risks posed to buildings and sports fields on the campus from 2011 flood.



Introduction

A flood risk analysis is performed on the main facilities at UQ for buildings and sports facilities using GIS.

Flood risk is given by the combination of likelihood and consequence; where: likelihood is given by flood frequency, and consequence is given by flood impacts. Below is risk matrix used for the assessment.

<... any additional explanation that explains the risk assessment in context for this assignment (max 100 words)>

According to the Environmental Impact Assessment, risk is predicated on concepts of hazard, chance and consequence. In this project, the qualitative assessment of likelihood is converted into the quantitative values, the elevation of the flooding. Likelihoods are categorised by flood recurrence intervals, such as one in twenty years, one in fifty years, and one in a hundred years, according to the BCC Fooldwise Report, the flood elevation is 3.7m, 5.5m, and 6.7m, respectively. For example, the same consequence with higher elevations is less flood occur, and the risk of flooding decreases. Details are displayed below (Table 1).

Table 1: Risk matrix. Likelihood is given by flood frequency occurrence as 1:20 year, 1:50 year or 1:100 year. Consequence is given by the area affected by the flood as being minor, significant or major by asset type.

		Consequence		
		Minor	Significant	Major
Likelihood	Likely 1:20 yr	Moderate	High	High
	Possible 1:50 yr	Low	Moderate	High
	Unlikely 1:100 yr	Low	Low	Moderate

Table 2: Consequences for facilities given % area affected by flood

Consequence →	Minor	Significant	Major
<i>Buildings Facilities % Affected</i>	>1%	>5%	>10%
<i>Sports Facilities % Affected</i>	>5%	>25%	>50%

Risk assessment for buildings

< ... complete table below and briefly describe (max. 100 words) your result, include supporting GIS maps, tables, charts, etc.>

Table 3: Risk assessment results of buildings

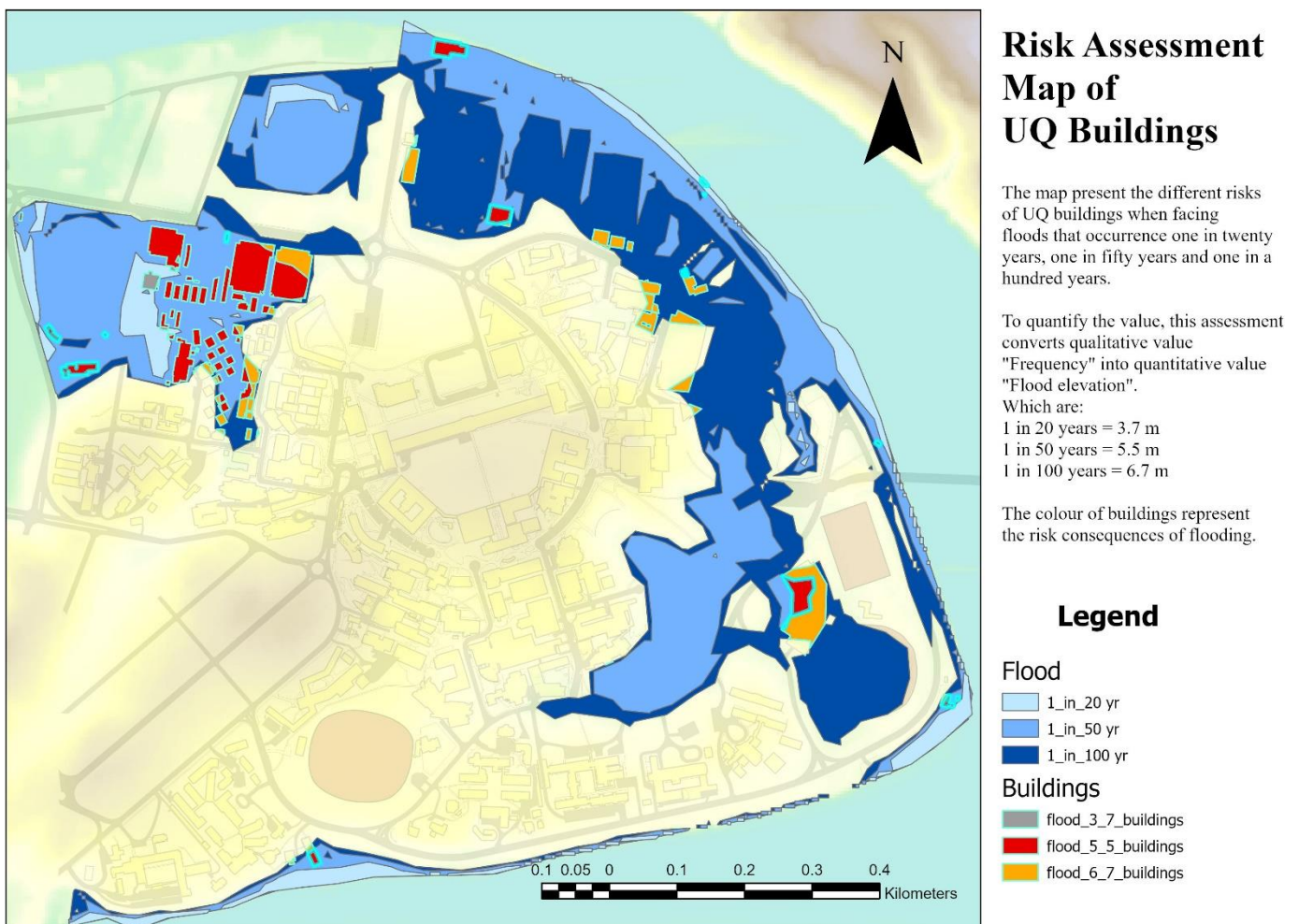
	Flood	Level (m)	Area (m ²)	% Affected	Conseq.	Likelihood	Risk
Buildings	1:20	3.7	756.419	~0.37%	None	Likely	None
	1:50	5.5	21,210.1	~10.2%	Major	Possible	High
	1:100	6.7	34,125.7	~16.5%	Major	Unlikely	Moderate

*The gross area of UQ buildings is 207,204.6m².

<...add GIS map output as figure and brief description on how to interpret the result>

Part of the UQ building will be affected by flooding. Due to the height of buildings, there is no risk of a 1-in-20-year flood; only a few buildings will be affected. When the flood height rises to 5.5m, exceeding the height of the first floor, the risk of 1-in-50-year flooding becomes high. With the decrease in frequency, there is a moderate risk of a 1-in-100-year flood; 16.5% of buildings will be affected. According to the map, floodwater invades the northeast and northwest parts of the campus from the riverbank, particularly buildings in the northwest.

Figure 1: Risk Assessment Map of UQ Buildings



Risk assessment for sports facilities

<... complete table below and briefly describe (max. 100 words) your result, include supporting GIS maps, tables, charts, etc. >

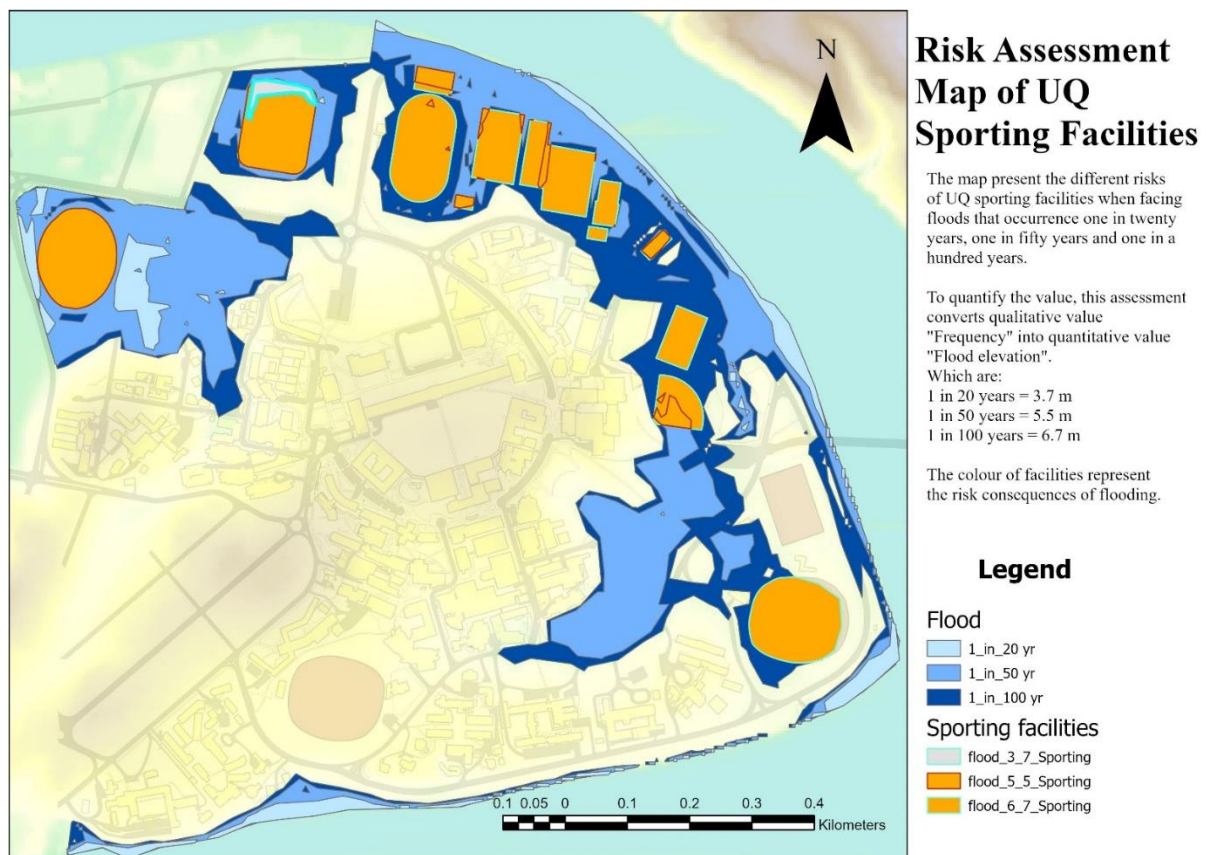
Table 4: Risk assessment results of sporting facilities

	Flood	Level (m)	Area (m ²)	% Affected	Conseq.	Likelihood	Risk
Sporting Facilities	1:20	3.7m	2207.73	~1.76%	None	Likely	None
	1:50	5.5m	37,567.1	~29.9%	Significant	Possible	Moderate
	1:100	6.7m	98,982.8	~78.8%	Major	Unlikely	Moderate

*The gross area of UQ sporting facilities is 125,643.6 m².

Majority of sporting facilities face flooding risk. According to the sporting facilities' depositions, all of them are located near the Brisbane River, which is easily affected by flooding. Besides, the sporting facilities' terrain mostly was flat and expansive; all of the sporting facilities will be damaged completely when it is a 1 in 100-year flood and will be damaged when it is a 1 in 50-year flood. Only one sporting facility will be damaged from a 1 in 20-year flood.

Figure 1: Risk Assessment Map of UQ Sporting Facilities



Discussion

<...briefly suggest management actions for results above and any limitations or assumptions you deem relevant to approach and the results...>

The floodwater invades the northeast and northwest regions of the campus from the riverbank. Hence, flood immigration should focus on the northeast and northwest riverbanks. Building additional drainage facilities can transfer floodwater when floods arrive; enhancing the density of vegetation can decelerate flood flow velocity.

For multi-story buildings with the risk of flooding can suspend the first floor or convert it into a parking lot. The single-story buildings with a flooding risk can be built with cheap materials. All those actions minimizes the impact of flooding on the building itself and facilitate cleanup or repair after flooding.

Sporting facilities can adopt permeable ground materials to facilitate post-flood cleanup and drying.

All management actions involve high costs and prolonged management periods for these existing buildings and facilities. Furthermore, all buildings are occupied with different functions; it is hard to renovate them.

Tolerable risk assessment

A field inspection of buildings was conducted for two purposes: i) an individual risk assessment of potential flood impacts on individual buildings, ii) judge if potential impacts are acceptable given actions may be taken to mitigate risk or intolerable where more structural interventions are required to bring situation to an acceptable risk.

Qld-SPP-Integrating-Interests-2021: A fit-for-purpose risk assessment is undertaken to identify and achieve an acceptable or tolerable level of risk for personal safety and property in natural hazard areas. mitigates the risks to people and property to an acceptable or tolerable level

acceptable – individuals and society can live with this risk without further action and accept any residual risk.

- tolerable – society can live with this risk but expect that as much as is reasonably practical should be done to reduce the risks further. Individuals may find this risk intolerable and choose to take their own steps, within reason, to make this risk acceptable.

- intolerable/unacceptable – individuals and society will not accept this risk and measures should be put in place to reduce risks to at least a tolerable level.

The table below identifies buildings and assessments for flood impacts based on:

- Likelihood of flooding in buildings having a significant or major consequence to first and (if applicable) second floors
- Proximity to the river (within 40 meters) where flood water movement is faster
- Assessment of tolerate risk with outcomes and comments on the assessment or possible measures to fix:
 - o Yes – acceptable to live with the risks as it has a negligible impact or simple practical measures may be taken to reduce impacts
 - o No – unacceptable risk and that require measures to mitigate risk.

Building ObjectID	Flr1_Flood	Flr2_Flood	Near_River	Tolerable	Comment
22	Possible	Not any	Yes	Yes	Higher likelihood, but a shed is cheap to fix.
252	Possible	Not any	No	No	Higher likelihood, that the first floor is mostly affected and the structure is made of glass and other fragile material
20	Possible	Not any	No	Yes	Higher likelihood, but mainly are pools, easy to refine
210	Unlikely	Not any	No	Yes	Lower likelihood, and the building was built with flood control measure
97	Possible	Not any	No	Yes	Higher likelihood, but parking lots are easy to refine from the flood as long as the park is car-free before the flood
117	Possible	Not anys	Yes	No	Higher likelihood and near the river, also the first floor is made of glass, but lacks protection
184	Possible	Not anys	No	Yes	Higher likelihood, but the affected first floor is a parking lot, easy to clean and refine after the flood
166	Possible	Not any	No	No	Higher likelihood, of only one floor, and the structure and materials are fragile, properties inside the building are expensive to compensate
159	Possible	Not any	No	Yes	Higher likelihood, that already have flood control measures and are cheap to fix
171	Possible	Not anys	No	Yes	Higher likelihood, but the affected first floor is a parking lot, easy to clean and refine after the flood
133	Possible	Not any	No	Yes	Higher likelihood, but fully enclosed structure
24	Unlikely	Not any	No	Yes	Lower likelihood and structure firm